

Cost-benefit Trade-off for Multi-UAV Assisted Sensor Data Collection in Jamming Environment

1st Yuan Bian

Air Force Engineering University
Xi'an, China
bianyuanlara@163.com

2nd Jianbo Hu

Air Force Engineering University
Xi'an, China
jian_bo_h@163.com

3rd Yingyang Wang

Air Force Engineering University
Xi'an, China
wangyingyang@outlook.com

4th Wenjie Liu

Air Force Engineering University
Xi'an, China
f254446635@163.com

5th Shuo Wang

Xidian University
Xi'an, China
shuowang@xidian.edu.cn

6th Peng Zhang*

Air Force Engineering University
Xi'an, China
wyrufu@163.com
*Corresponding author

Abstract—This article investigates the trade-off among average uplink rate, propulsion energy consumption, and sensor energy efficiency in multi-unmanned aerial vehicle (UAV) data collection systems, with a particular focus on obstacle avoidance and collision prevention. A constrained multi-variable optimization problem is formulated, incorporating real-time constraints such as sensor upload rates, obstacle avoidance, and collision-free trajectory requirements. To solve this, a two-layer optimization framework is proposed, integrating block coordinate descent (BCD) and serial coordination to decompose the problem into tractable subproblems: sensor scheduling, power allocation, and trajectory planning. Notably, a penalty-based successive convex approximation (SCA) method is employed to enforce collision-free constraints during trajectory optimization, ensuring safe in cluttered environments. Simulation results demonstrate that adjusting weight parameters allows the algorithm to dynamically prioritize cost-benefit trade-offs while robustly maintaining obstacle avoidance and collision prevention, even in scalable multi-UAV scenarios.

Index Terms—data collection, multi-UAV trajectory, anti-jamming, trade-off, penalty-based method, obstacle avoidance

I. INTRODUCTION

UAV, with their low cost, high mobility, and flexible deployment, demonstrate significant value in various applications [1]. Their mobility yields two benefits via trajectory optimization: (a) improved channel quality by reducing transmission distance, (b) jammer/obstacle avoidance. This transforms spatial degrees of freedom into improved quality of communication service [2]. Energy consumption remains a critical bottleneck for UAV-assisted internet of things (IoT); therefore, existing research prioritizes minimizing energy or maximizing efficiency through joint trajectory-communication optimization. Furthermore, broadcast channels are vulnerable to jamming, which disrupts transmissions, degrades performance, and compromises energy optimization, as UAVs require additional energy to counteract interference [3]. Hence, effective trajectory design and resource allocation require joint optimization of the energy-rate trade-off under the jamming attack to ensure

reliable IoT service delivery.

[4] minimized UAV and sensor node energy via joint 3D trajectory and data scheduling using a new propulsion model. [5] minimized weighted UAV/sensor energy through clustering and routing in multi-UAV systems. [6] derived Pareto-optimal UAV-GT (ground terminal) energy trade-offs for circular/linear flight patterns in ground-UAV systems. [7] minimized weighted node/UAV energy with secure 3D trajectory optimization under eavesdropping. [8] maximized secrecy rate while minimizing user energy under latency constraints. [9] balanced task time and energy by co-designing trajectories/hover positions/speeds via ant colony optimization.

In the context of joint trajectory optimization and communication resource allocation for UAV-assisted information collection systems, existing studies [4]- [9] have made progress in incorporating broader operational dimensions such as sensor energy efficiency and temporal resource management. However, two critical research gaps persist: First, works [4]- [6] overlook essential performance indicators (e.g., throughput maximization) that directly impact mission effectiveness, creating a misalignment between theoretical optimization objectives and practical task requirements. Second, while [7]- [9] introduce multi-UAV coordination frameworks, their models rely on idealized propagation assumptions that neglect real-world constraints such as obstacle avoidance mechanisms and anti-jamming capabilities, thereby limiting algorithmic robustness in non-ideal environments. Although [5] and [9] address multi-UAV data collection scenarios, their optimization formulations fail to integrate UAV kinematic constraints (e.g., velocity/acceleration limits), which are fundamental for trajectory feasibility in physical space. The cumulative effect of these limitations is a persistent gap between theoretical models and operational realities, particularly in complex scenarios requiring collision-free coordination among UAVs.

In this paper, we delve into the investigation of fixed-

wing UAV-assisted wireless data collection within jamming environments. The primary objective is to maximize a weighted sum encompassing the average upload rate, UAV propulsion energy, and sensor transmission energy. This is achieved through the joint optimization of sensor scheduling, sensor power allocation, and UAV trajectory utilizing the BCD method. Specifically, we propose an optimization approach for multi-UAV trajectory based on penalty functions, employing the SCA technique. This approach is designed to address the coupled constraints associated with obstacle avoidance and UAV-to-UAV collision prevention.

II. SYSTEM MODEL

We consider a multi-UAV-assisted uplink data collection system where ground sensors ($\mathcal{S} \triangleq \{1, 2, \dots, S\}$) upload data to UAVs ($\mathcal{U} \triangleq \{1, 2, \dots, U\}$) among jammers ($\mathcal{J} \triangleq \{1, 2, \dots, J\}$) and obstacles ($\mathcal{O} \triangleq \{1, 2, \dots, O\}$). The coordinates for UAV $u \in \mathcal{U}$, sensor $s \in \mathcal{S}$, and jammer $j \in \mathcal{J}$ are $\mathbf{q}_u = (x_u, y_u, H)$, $\mathbf{q}_s = (x_s, y_s)$, and $\mathbf{q}_j = (x_j, y_j)$, respectively. The UAV flight duration T is discretized into N equal time slots, each of duration $\delta_t = T/N$. Assuming δ_t is sufficiently small, the UAV's location and speed remain approximately constant within each slot. Under this discretization, the following physical constraints are imposed:

$$\|\mathbf{v}_u[n]\| \leq v_{\max}, \forall n = 1, \dots, N \quad (1)$$

$$\|\mathbf{v}_u[n]\| \geq v_{\min}, \forall n = 1, \dots, N \quad (2)$$

$$\|\mathbf{a}_u[n]\| \leq a_{\max}, \quad \forall n = 1, \dots, N \quad (3)$$

$$\mathbf{v}_u[0] = \mathbf{v}_0, \quad \mathbf{v}_u[N] = \mathbf{v}_N \quad (4)$$

$$\mathbf{q}_u[0] = \mathbf{q}_0, \quad \mathbf{q}_u[N] = \mathbf{q}_N \quad (5)$$

$$\mathbf{v}_u[n] = \mathbf{v}_u[n-1] + \mathbf{a}_u[n]\delta_t, \quad \forall n = 2, \dots, N \quad (6)$$

$$\mathbf{q}_u[n] = \mathbf{q}_u[n-1] + \mathbf{v}_u[n]\delta_t + \frac{1}{2}\mathbf{a}_u[n]\delta_t^2, \quad \forall n = 2, \dots, N \quad (7)$$

$$\|\mathbf{q}_u[n] - \mathbf{q}_{u'}[n]\|_2 \geq d_{\text{safe}}, \quad \forall n = 1, \dots, N \quad (8)$$

The UAV faces operational constraints: (1)-(2) set speed bounds; (3) limits acceleration; (4)-(5) govern launch/recovery velocity and trajectory; (6)-(7) regulate in-flight velocity and trajectory; (8) guarantees that the distance between UAVs remains greater than d_{safe} at all times.

For UAV obstacle avoidance in data collection, consider O obstacles indexed by $\mathcal{O} \triangleq 1, \dots, O$. Each is modeled as an infinitely tall cylinder with ground-projected center $\mathbf{q}_o = (x_o, y_o)$ and radius r_o . To avoid collisions, the UAV must keep an avoidance distance above r_o , leading to:

$$\|\mathbf{q}_u[n] - \mathbf{q}_o[n]\|_2 \geq r_o, n = 1, \dots, N, o = 1, 2, \dots, O \quad (9)$$

Based on the free-space path loss model [10], the channel power gain of sensor-UAV link and the channel power gain of jammer-UAV link can be modeled as follows.

$$h_{ku}[n] = \beta_0 d_{ku}^{-2}[n] = \frac{\beta_0}{\|\mathbf{q}_k - \mathbf{q}_u[n]\|^2} \quad k = s, j \quad (10)$$

where β_0 represents the channel power gain at 1m. Therefore, the achievable transmission rate (bits per second) from the sensor node s to the UAV u at time slot n is

$$R_{su}[n] = B \log_2 \left(1 + \frac{P_s[n] h_{su}[n]}{\sum_{j=1}^J P_j h_{ju}[n] + \sigma^2} \right) \quad (11)$$

where P_j is the jammer power, σ^2 is the Gaussian white noise power, and $P_s[n]$ is the sensor transmit power restricted by (12) (maximum power).

$$0 \leq P_s[n] \leq P_{\max} \quad (12)$$

Let $x_{su}[n] \in [0, 1]$ be the schedule variable where $x_{su}[n] = 1$ indicates that sensor s transmits to UAV u at time n . under Time Division Multiple Access (TDMA) protocol, and each UAV serves at most one sensor per time slot and each sensor uploads data to at most one UAV.

$$\sum_{u=1}^U x_{su}[n] \leq 1 \quad \forall n, s \quad (13)$$

$$\sum_{s=1}^S x_{su}[n] \leq 1 \quad \forall u, n \quad (14)$$

To guarantee each sensor's minimum required average data transmission rate $R_{\min}$, we have

$$\frac{1}{N} \sum_{u=1}^U \sum_{n=1}^N x_{su}[n] R_{su}[n] \delta_t \geq R_{\min} \quad \forall s \quad (15)$$

System energy includes sensor transmission and UAV propulsion. We only consider sensor data transmission energy for sensor energy consumption, ignoring circuit-level usage.

$$E_s = \sum_{s=1}^S \sum_{n=1}^N P_s[n] \quad (16)$$

Based on the propulsion model in [11], the power consumption of fixed-wing UAV u is:

$$E_u = \sum_{n=1}^N \left(c_1 \|\mathbf{v}[n]\|^3 + \frac{c_2}{\|\mathbf{v}[n]\|} \left(1 + \frac{\|\mathbf{a}[n]\|^2}{g^2} \right) \right) \quad (17)$$

where c_1 and c_2 are the associated aerodynamic parameters, and g is the gravitational acceleration.

Expressions (2) and (16) reveal nonlinear coupling between sensor energy consumption E_s , communication variables ($P_s[n]$, $h_{su}[n]$, $h_{ju}[n]$, $x_{su}[n]$), and UAV trajectory. While shorter $x_{su}[n]$ reduces energy by limiting transmission time, this requires optimization of the UAV trajectory to maintain performance through reduced communication distances. This trade-off requires joint optimization of power allocation ($\mathbf{P} \triangleq \{P_s[n]\}$), schedule $\mathbf{X} \triangleq \{x_{su}[n]\}$, and trajectory $\mathbf{Q} \triangleq \{\mathbf{Q}_u[n]\}$ to balance average rate, sensor energy consumption, UAV propulsion energy consumption.

To quantify the energy-rate trade-off, we define: revenue per bit/s λ_1 , sensor energy cost per Joule λ_2 , and UAV propulsion

cost per Joule λ_3 . These collectively establish a unified cost-benefit framework, yielding:

$$C = \lambda_1 \left(\frac{1}{T} \sum_{s=1}^S \sum_{u=1}^U \sum_{n=1}^N x_{su}[n] R_{su}[n] \right) - \lambda_2 E_s - \lambda_3 \sum_{u=1}^U E_u \quad (18)$$

Based on the above, the original optimization problem can be formally formulated as follows:

$$(P1) \quad \max_{\mathbf{Q}, \mathbf{X}, \mathbf{P}} \quad C(\mathbf{Q}, \mathbf{X}, \mathbf{P}) \quad (19)$$

s.t. (1) – (9), (12) – (16)

Problem (P1) is non-convex and intractable due to a non-concave objective (20), non-convex constraints (2), (8), (9), (16), and strongly coupled variables $\mathbf{Q}$, $\mathbf{X}$, and $\mathbf{P}$.

III. PROPOSED SOLUTION

In this section, the BCD algorithm is utilized to decompose the original problem (P1) into three sequentially optimized subproblems: sensor schedule $\mathbf{X}$, power allocation $\mathbf{P}$, and multi-UAV trajectories $\mathbf{Q}$ optimization.

A. Sensor schedule optimization

In this subsection, with fixed sensor power $\mathbf{P}$ and multi-UAV trajectories $\mathbf{Q}$, the energy terms E_s and E_u become constant, reducing the cost-benefit function C to $C_1 = \frac{1}{T} \sum_{s=1}^S \sum_{u=1}^U \sum_{n=1}^N x_{su}[n] R_{su}[n]$ and transforming (P1) into (P2.1):

$$(P2.1) \quad \max_{\mathbf{X}} \quad C_1(\mathbf{X}) \quad (20)$$

s.t. (14) – (16)

Since $x_{su}[n]$ is binary, we apply linear relaxation by replacing it with a continuous variable constrained to [0,1]:

$$0 \leq x_{su}[n] \leq 1 \quad (21)$$

After the above transformation, the optimization problem (P2.1) is transformed into (P2.2):

$$(P2.2) \quad \max_{\mathbf{X}} \quad C_1(\mathbf{X}) \quad (22)$$

s.t. (14) - (16), (21)

Given the linearity of C_1 and the constraints, (P2.2) is a standard linear programming (LP) problem solvable via tools like CVX.

B. Sensor Power Allocation

With fixed $\mathbf{X}$ and $\mathbf{Q}$, E_u is deterministic, reducing cost function C to $C_2 = \lambda_1 \left(\frac{1}{T} \sum_{s=1}^S \sum_{u=1}^U \sum_{n=1}^N x_{su}[n] R_{su}[n] \right) - \lambda_2 E_s$ and reformulating (P1) as (P3):

$$(P3) \quad \max_{\mathbf{P}} \quad C_2(\mathbf{P}) \quad (23)$$

s.t. (12), (16)

Problem (P3) is convex since its objective is concave in $\mathbf{P}$, while constraint (16) has a concave left hand side (LHS) and constant right hand side (RHS), making it solvable directly via convex solvers.

C. Multi-UAV Trajectory Optimization

Under ideal communications with unlimited pairwise connectivity, given fixed $\mathbf{P}$ and $\mathbf{X}$, we optimize U UAVs trajectories $\mathbf{Q} \triangleq \{\mathbf{Q}_1[n], \dots, \mathbf{Q}_U[n]\}, u \in U$. The objective is seeking optimal trajectories $\mathbf{Q}^*[n] \triangleq \{\mathbf{Q}_1^*[n], \dots, \mathbf{Q}_U^*[n]\}$ satisfying: collision avoidance, obstacle evasion, kinematic constraints, QoS, and initial/terminal positions.

To achieve this, a sequential decoupling strategy is employed. This strategy involves solving the trajectory optimization problem for each UAV sequentially, treating the trajectories of the other UAVs as fixed during each individual optimization step. Specifically, the process of multi-UAV trajectory optimization proceeds as follows.

- **Sequential Optimization:** For each UAV u , the trajectories of all other UAVs ($u' \in U \setminus \{u\}$) are held fixed. The optimization then centers on determining the optimal trajectory $\mathbf{Q}_u^*[n]$ for the current UAV using the SCA technique.

- **Anti-Collision Constraints transformation:** The optimal trajectory of the current UAV u is treated as a known parameter and incorporated into the anti-collision constraint for the subsequent $(u+1)$ -th UAV's trajectory optimization. This approach systematically resolves spatial conflicts by leveraging inter-UAV trajectory dependencies.

Then, taking UAV u as an example, when the cost function C transitions to $C_3 = \lambda_1 \left(\frac{1}{T} \sum_{s=1}^S \sum_{u=1}^U \sum_{n=1}^N x_{su}[n] R_{su}[n] \right) - \lambda_3 E_u$, the optimization problem (P1) can be decoupled and converted to (P4):

$$(P4.1) \quad \max_{\mathbf{Q}_u} \quad C_3(\mathbf{Q}, \mathbf{X}, \mathbf{P}) \quad (24)$$

s.t. (1) – (9), (16)

The objective function (24) is non-concave with respect to $\mathbf{q}_u[n]$, and constraints (2), (8), (9), and (15) are non-convex in $\mathbf{q}_u[n]$. Therefore, addressing these non-convex terms is essential for finding a solution.

1) *Transformation of the objective function $R_{su}[n]$:* For $R_{su}[n]$, we have:

$$R_{su}[n] = \hat{R}_{su}[n] - \bar{R}_{su}[n] \quad (25)$$

where

$$\hat{R}_{su}[n] = \log_2 \left(1 + \sum_{j=1}^J \frac{P_j \beta_0 / \sigma^2}{(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 + H^2)} + \frac{P_s[n] \beta_0 / \sigma^2}{(\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 + H^2)} \right)$$

$$\bar{R}_{su}[n] = \log_2 \left(1 + \sum_{j=1}^J \frac{P_j \beta_0 / \sigma^2}{(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 + H^2)} \right)$$

Given the convex $\hat{R}_{su}[n]$ in $\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2$ and $\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2$, its first-order Taylor expansion at r -iteration yields global lower bound $\hat{R}_{su}^{lb}[n]$:

$$\begin{aligned} \hat{R}_{su}[n] &\geq A_{su}^r[n] + B_{su}^r[n](\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 - \|\mathbf{q}_s - \mathbf{q}_u^r[n]\|^2) \\ &\quad + C_{ju}^r[n](\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 - \|\mathbf{q}_j - \mathbf{q}_u^r[n]\|^2) \\ &\triangleq \hat{R}_{su}^{lb}[n] \end{aligned} \quad (26)$$

where

$$A_{su}^r[n] = \log_2(I_{su}^r[n])$$

$$I_{su}^r[n] = 1 + \sum_{j=1}^J \frac{P_j \beta_0 / \sigma^2}{(\|\mathbf{q}_j - \mathbf{q}_u^r[n]\|^2 + H^2)} + \frac{P_s[n] \beta_0 / \sigma^2}{(\|\mathbf{q}_s - \mathbf{q}_u^r[n]\|^2 + H^2)}$$

$$B_{su}^r[n] = -\frac{P_s[n] \beta_0}{\sigma^2 I_{su}^r[n] \ln 2} \frac{1}{(\|\mathbf{q}_s - \mathbf{q}_u^r[n]\|^2 + H^2)^2}$$

$$C_{ju}^r[n] = -\frac{P_j \beta_0}{\sigma^2 I_{su}^r[n] \ln 2} \frac{1}{(\|\mathbf{q}_j - \mathbf{q}_u^r[n]\|^2 + H^2)^2}$$

Introducing relaxation variables $\{d_j[n]\}$, we have:

$$\frac{P_j \beta_0 / \sigma^2}{(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 + H^2)} \leq e^{d_j[n]} \quad (27)$$

With (26) and (27), we have

$$R_{su}[n] \geq \hat{R}_{su}^{lb}[n] - \log_2(1 + \sum_{j=1}^J e^{d_j[n]}) \triangleq R_{su}^{lb}[n] \quad (28)$$

2) *Transformation of the constraints:* Then, the first-order Taylor expansion is conducted for $\mathbf{v}_u[n]$, constraint (2) is reformulated as follows.

$$\begin{aligned} \|\mathbf{v}_u[n]\|^2 &\geq \|\mathbf{v}_u^r[n]\|^2 + 2\mathbf{v}_u^r[n]^T(\mathbf{v}_u[n] - \mathbf{v}_u^r[n]) \\ &\triangleq \varphi(\mathbf{v}_u[n]; \mathbf{v}_u^r[n]) \\ &\geq v_{min} \end{aligned} \quad (29)$$

The non-convex item E_u is reformulated as (30) by introducing the relaxing variable $\mu[n]$.

$$\mu^2[n] \leq \varphi(\mathbf{v}_u[n]; \mathbf{v}_u^r[n]) \quad (30)$$

$$E_u \leq \sum_{n=1}^N (c_1 \|\mathbf{v}[n]\|^3 + \frac{c_2}{\|\mu^2[n]\|} (1 + \frac{\|\mathbf{a}[n]\|^2}{g^2})) \triangleq E_u^{ub} \quad (31)$$

Non-convex constraint (8) is linearized via first-order Taylor expansion at r -th iteration point $\mathbf{q}_u^r[n]$ in a convex form:

$$\begin{aligned} &\|\mathbf{q}_u^r[n] - \mathbf{q}_u^r[n]\|_2 + \\ &\frac{(\mathbf{q}_u^r[n] - \mathbf{q}_u^r[n])^T}{\|\mathbf{q}_u^r[n] - \mathbf{q}_u^r[n]\|} (\mathbf{q}_u[n] - \mathbf{q}_u^r[n]) \geq d_{safe} \end{aligned} \quad (32)$$

For the obstacle avoidance constraint (9), a first-order Taylor expansion is applied to derive the following.

$$\begin{aligned} \|\mathbf{q}_u[n] - \mathbf{q}_o\|_2 &\geq \|\mathbf{q}_u^r[n] - \mathbf{q}_o\|_2 \\ &+ \frac{(\mathbf{q}_u^r[n] - \mathbf{q}_o)^T}{\|\mathbf{q}_u^r[n] - \mathbf{q}_o\|} (\mathbf{q}_u[n] - \mathbf{q}_u^r[n]) \triangleq \varphi(\mathbf{q}_u[n]; \mathbf{q}_u^r[n]) \end{aligned} \quad (33)$$

Thus, non-convex constraint (9) is transformed into (34):

$$\varphi(\mathbf{q}_u[n]; \mathbf{q}_u^r[n]) \geq r_o \quad (34)$$

And the constraint (15) can be transformed into the following form based on (28).

$$\frac{1}{N} \sum_{u=1}^U \sum_{n=1}^N x_{su}[n] R_{su}^{lb}[n] \delta_t \geq R_{min} \quad \forall s \in \mathcal{S} \quad (35)$$

In conclusion, the non-convex optimization problem (P4.1) is reformulated as the convex problem (P4.2):

$$C_4 = \lambda_1 \left(\frac{1}{T} \sum_{s=1}^S \sum_{u=1}^U \sum_{n=1}^N x_{su}[n] R_{su}^{lb}[n] \right) - \lambda_3 E_u^{ub} \quad (36)$$

$$\begin{aligned} &\text{(P4.2)} \quad \max_{\mathbf{Q}_u, \{\mu[n]\}, \{ju[n]\}} C_4(\mathbf{Q}_u) \\ &\text{s.t.} \quad (1), (3) - (6), (30) - (35) \end{aligned} \quad (37)$$

Given the non-convex nature of collision avoidance con-

Algorithm 1 Solving (P4.3) of Multi-UAV Trajectory Optimization

Require: Trajectory matrix $\mathbf{Q}^1[n]$, penalty coefficient C'_1 , maximum iterations K_{max} , number of UAVs U , tolerance ε .

Step 1: Initialize $k = 1, u = 1$.

Step 2: While $u < U$:

Step 3: Augment the objective function to form $\tilde{f}(\mathbf{Q}_u)$ as (38) shows.

Step 4: Solve (P4.3) using SCA to obtain $\mathbf{Q}_u^{k+1}[n]$.

Step 5: Adjust C'_k into C'_{k+1} dynamically according to (40)

Step 6: Check Convergence:

if $V^{(k+1)}(\mathbf{q}_u[n]) \leq \varepsilon$ **and** $|\tilde{f}(\mathbf{q}_u^{k+1}[n]) - \tilde{f}(\mathbf{q}_u^k[n])|$ is negligible, or $k \geq K_{max}$ **then**

Accept $\mathbf{Q}_u^{k+1}[n]$ as the optimal trajectory $\mathbf{Q}_u^*[n]$ for UAV u and proceed to **Step 7**.

else

Increment k to $k + 1$ and return to **Step 3**.

Step 7: Update UAV index and return to **Step 2**.

$$u = u + 1$$

Step 8: Output optimal trajectories $\mathbf{Q}^*[n]$ for all UAVs.

straints and the possibility that problem (P4.2) may become infeasible owing to iterative conservative convex approximations, we adopt a penalty function method as described below.

3) *Trajectory Optimization based on penalty function:* The hard constraint (33) is embedded into the objective function $C_4(\mathbf{Q}_u)$ via the penalty function $V^{(k)}(\mathbf{q}_u[n]; \mathbf{q}_u^k[n])$, yielding an augmented objective function $\tilde{f}(\mathbf{Q}_u)$. This reformulates the optimization problem as (P4.3).

$$\tilde{f}(\mathbf{Q}_u) = C_4(\mathbf{Q}_u) + C'_k \sum_{n=1}^N V^{(k)}(\mathbf{q}_u[n]; \mathbf{q}_u^k[n]) \quad (38)$$

$$V^{(k)}(\mathbf{q}_u[n]; \mathbf{q}_u^k[n]) = \min\{0, \varphi(\mathbf{q}_u[n]; \mathbf{q}_u^k[n]) - r_o\} = \begin{cases} 0, & \|\mathbf{q}_u[n] - \mathbf{q}_o\|_2 > r_o \\ \varphi(\mathbf{q}_u[n]; \mathbf{q}_u^k[n]) - r_o, & \|\mathbf{q}_u[n] - \mathbf{q}_o\|_2 \leq r_o \end{cases} \quad (39)$$

$$C'_k = \begin{cases} \gamma C'_{k-1} & \text{if } \mathbf{V}^{(k)}(\mathbf{q}_u^k[n]) \leq \mathbf{V}^{(k-1)}(\mathbf{q}_u^{k-1}[n]) \\ C'_{k-1} & \text{if } \mathbf{V}^{(k)}(\mathbf{q}_u^k[n]) > \mathbf{V}^{(k-1)}(\mathbf{q}_u^{k-1}[n]) \end{cases} \quad (40)$$

where $\gamma > 1$ is a growth factor, C'_k is the penalty factor that governs the penalty strength for constraint violations in the k -th iteration, and the penalty function $V^{(k)}(\mathbf{q}_u[n]; \mathbf{q}_u^k[n])$ is defined in (39).

$$\begin{aligned} \text{(P4.3)} \quad & \max \tilde{f}(\mathbf{Q}_u) \\ \text{s.t.} \quad & (1), (3) - (6), (30) - (33), (36), (39) \end{aligned} \quad (41)$$

The steps for solving problem (P4.3) are outlined as follows.

D. Overall Algorithm

In this section, we propose Algorithm 2 to compute a suboptimal solution for (P1) by alternating optimization of $\mathbf{Q}$, $\mathbf{X}$, and $\mathbf{P}$ until convergence, as detailed below. Subproblem (P2.2) is a LP solvable by the interior point method with complexity $O(\sqrt{USN})$, where U , S , and N are the numbers of UAVs, sensors, and time slots, respectively. Similarly, P(3) is a convex problem with complexity $O(L_1 SN^{3.5})$, where L_1 is the iteration number. For P(4.3), the complexity is $O(UL_u SN^{3.5})$, where L_u is the iteration number for UAVs' trajectories optimization. The overall complexity of Algorithm 2 is $O(L(\sqrt{USN} + L_1 SN^{3.5} + UL_u SN^{3.5}))$.

Algorithm 2 Solve P(1) via Iterative Optimization

Require: Initialize $\mathbf{Q}^1, \mathbf{P}^1, \mathbf{X}^1$; $l \leftarrow 1$
repeat
 Update $\mathbf{X}$ by solving P(2.2) with current $\mathbf{Q}^l, \mathbf{P}^l$
 Update $\mathbf{P}$ by solving P(3) with current $\mathbf{Q}^l, \mathbf{X}^{l+1}$
 Update $\mathbf{Q}$ by solving P(4.3) with latest $\mathbf{X}^{l+1}, \mathbf{P}^{l+1}$
 $l \leftarrow l + 1$
until Convergence = 0

IV. NUMERICAL RESULTS

We consider a system model composed of $M = 4$ UAVs operating at a fixed altitude of $H = 150$ m, and $S = 10$ sensors distributed in a 700×1200 m² area. Key parameters are defined as follows: $P_{max} = 1$ W, $P_{peak} = 0.7$ W, $\beta_0 = -60$ dB, $\sigma_0^2 = -110$ dBm, $V_{min} = 3$ m/s and $V_{max} = 150$ m/s, $a_{max} = 5$ m/s², $c_1 = 9.26 \times 10^4$, $c_2 = 2250$, $N = 40$, $\delta_t = 1$ s, $\gamma = 5$, $C'_1 = 10$.

In Fig.1(a), as the weight of average upload rate λ_1 increases with $\lambda_2 = 100$ and $\lambda_3 = 1$, the sensor's energy consumption E_s rises significantly. It grows gradually at lower upload rates but accelerates at higher ones, likely due to the increased energy required for uploading larger data

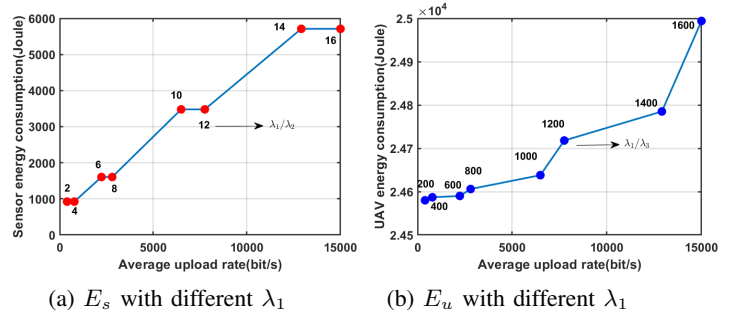
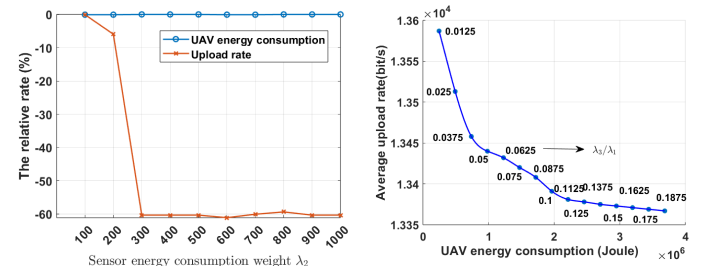


Fig. 1: Trade-off with different λ_1

volumes. Similarly, the UAV's energy consumption E_u also rises with λ_1 in Fig.1(b), increasing gradually at low to moderate rates before spiking at high rates, possibly because the UAV must adjust its trajectory to stay close to the sensors and maintain high upload rates.

As depicted in Fig.2(a), when the sensor energy consumption weight λ_2 is increased with fixed $\lambda_1 = 800$ and $\lambda_3 = 1$, the average upload rate may decrease. However, the UAV's energy consumption E_u remains unaffected. Nonetheless, once λ_2 surpasses a certain threshold—indicating that the system must reduce sensor power usage under prevailing environmental constraints—the average upload rate will decline to accommodate the increased emphasis on λ_2 . Currently, the UAV can adjust its trajectory to stabilize the upload rate and mitigate fluctuations.



(a) E_u and R_{su} with different λ_2 (b) R_{su} with different λ_3

Fig. 2: Trade-off with different λ_2 and λ_3

In Fig.2(b), as the weight assigned to UAV energy consumption λ_3 increases from 10 to 150 with $\lambda_1 = 800$ and $\lambda_2 = 1$, the upload rate generally shows a downward trend, indicating that a higher weight λ_3 may lead the optimization to focus more on reducing UAV energy consumption, thus necessitating a reduction in the upload rate.

As illustrated in Fig.3, it can also be observed that the 2-D UAV's trajectory under Strategy 1 with $\frac{\lambda_3}{\lambda_1} = 0.1875$

and $\lambda_3 = 150$ demonstrates a more pronounced curvature compared to that under Strategy 2 with $\frac{\lambda_3}{\lambda_1} = 0.0125$ and $\lambda_3 = 10$. This indicates that the trajectory is being optimized in a direction that enhances energy efficiency. In essence, the ratio of λ_3 to λ_1 serves as an indicator of the system's energy efficiency; the closer this ratio is to 1, the more efficiently the UAV's energy is utilized.

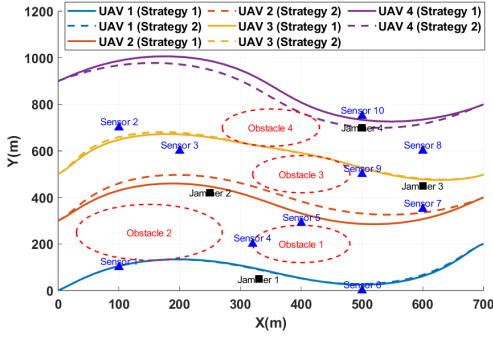


Fig. 3: 2-D Trajectory comparison

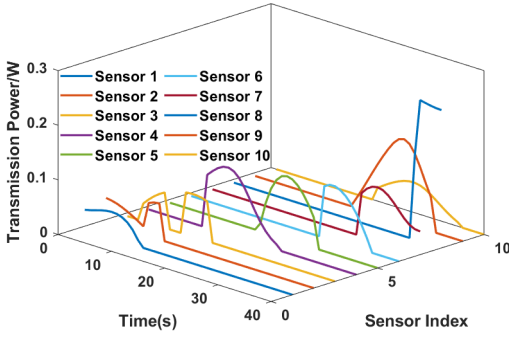


Fig. 4: Sensor schedule

Fig.4 shows the dynamic power allocation and schedule of sensors with $\lambda_1 = 800$, $\lambda_2 = 1000$ and $\lambda_3 = 1$. It can be observed that the sensor remains inactive for saving energy (with zero power consumption) until it establishes a connection with a specific UAV and begins uploading data.

We then compare the proposed scheme ($\lambda_1 = 800, \lambda_2 = 800, \lambda_3 = 1$) with two cases as in Table I. **Case 1:** rate-maximization, this scheme focuses on the only rate maximization, while other settings are same with the proposed scheme. **Case 2:** rate-sensor energy balance, this scheme focuses on the rate-sensor energy trade-off with the ratio of 1:1.

In practice, the proposed algorithm makes a balance among the three indexes, which can be flexibly configured to prioritize rate maximization, rate-sensor energy balance, energy efficiency, or minimum energy consumption by appropriately tuning the parameters λ_1 , λ_2 and λ_3 . This adaptability makes the algorithm suitable for diverse operational scenarios.

TABLE I: Comparison of different cases

	$R_{ave}(\text{bit/s})$	$E_u(\text{J})$	$E_s(\text{J})$
Case 1	19.3568	3.7358e+04	276.3579
Case 2	1.8667	3.7985e+04	7.8019
Proposed	1.9576	2.4582e+04	8.9269

V. CONCLUSION

This paper proposes a weighted sum maximization framework for uplink rate and energy consumption of UAV, sensors in multi-UAV-assisted data collection, addressing constrained optimization of sensor schedule, power allocation, and UAVs trajectories. The BCD-SCA algorithm is developed to alternately optimize these variables, with a penalty-based convex optimization framework introduced for obstacle avoidance Multi-UAV trajectory optimization. Simulations demonstrate that adjusting the weight parameters optimizes the cost-benefit trade-off, enabling obstacle avoidance while preserving scalability and adaptability to diverse operational scenarios. Future research will use parallel optimization to reduce complexity as UAV numbers grow, improving cost-effectiveness.

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